Seat No:_____ **Enrollment No:_____**

School of Computer Applications
Department of Master of Computer Applications

University Examination Summer – 2023

Subject Code: 40110201 **Date: 10/04/2023**

Subject Name: Data Structures (DS)

Duration: 02:30 Hours **Total Marks: 70**

Q1. Answer The Following Questions. [Draw suitable diagram whenever required]

I. What is a root node? (2)

II. Define: Algorithm (2)

III. What is complete binary tree? (2)

IV. Define: Data structures (2)

V. What is Hashing? (2)

VI. Define: Isolated and Pendent vertex (2)

VII. List out difference between linked list and array. (2)

Q2. Answer The Following Questions.(Attempt any four)

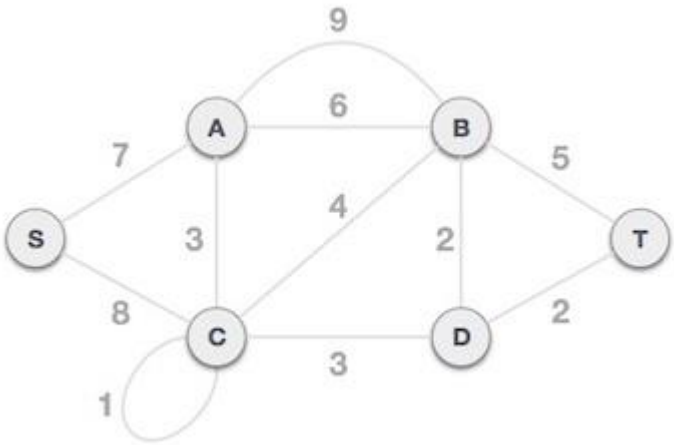
I. Consider an array X [-15…….10, 15……40] requires one byte of storage. If base address is 1500 then determine the address of X [15][20]. Assume that the elements are stored in row major order. (3)

II. Construct a Binary Search Tree by inserting the following sequence of numbers (3)

10,12,5,4,20,8,7,15 and 13

III. Explain classification of data structure with example. (3)

IV. Consider following graph. Find minimum spanning tree using Kruskal's algorithm . (3)



V. What is Stack? Write an algorithm for inserting an element into Stack. (3)

Q3. Answer The Following Questions. (Attempt any four)

I. What is Linked list? Write an algorithm to delete an element at last position in singly linked list. (3)

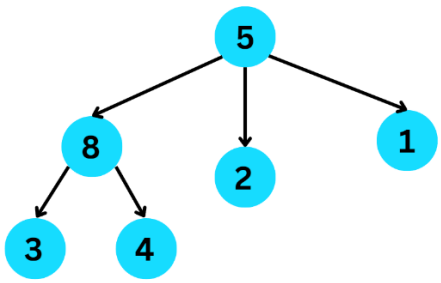
II. Explain operations performed on different data structures. (3)

III. Convert following expression from infix to postfix using stack. (3)

(A – (B / C + (D % E * F) / G)* H)

IV. Sort the numbers using Shell sort: 71, 7, 28, 22, 36, 12, 15, 22, 4 (Show every pass) (3)

V. Convert following general tree into binary tree. (3)



VI. Evaluate following postfix expression using stack. (3)

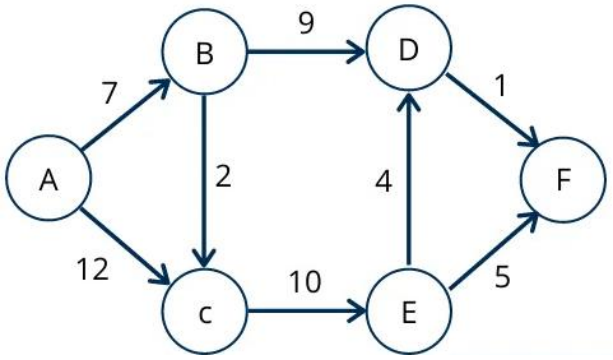
5,6,2,+,*,12,4,/,-

Q4. Answer The Following Questions. (Attempt any four)

I. What is Queue? List out types of Queue and write an algorithm to insert an element in the circular Queue. (4)

II. List out the ways to represent the Graph and explain Graph representation using Adjacency Matrix with suitable example. (4)

III. Find shortest path of following graph using Dijkstra's algorithm, consider A as source node. (4)



IV. Arrange following elements in sorted order using Radix sort. (4)
101, 45,543,233,321, 833,432 (Show every pass).

V. Write and explain Binary Search algorithm with suitable example. (4)

VI. Construct AVL tree for the following sequence of numbers. (4)

50 , 20 , 60 , 10 , 8 , 15 , 32 , 46 , 11 , 48

Q5. Answer The Following Questions. (Attempt any four)

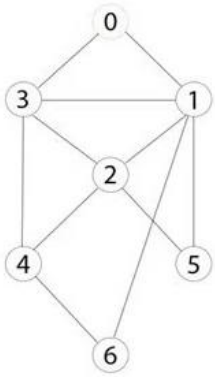
I. What is Hashing? Explain any two collision resolution techniques with suitable example. (4)

II. Define: 1) Complete Binary Tree 2) Sparse Matrix 3) Degree of a vertex 4) Minimum Spanning Tree (4)

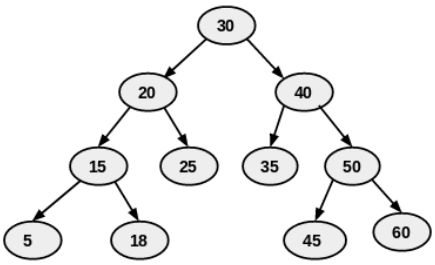
III. Explain time complexity and space complexity of an algorithm. (4)

IV. Apply insertion sort on the following data: 44,33,11,55,77,90,40,60 (Show every pass) (4)

V. Explain the ways to traverse a Graph. Find out BFS traversal for the following Graph. (consider 0 as the starting vertex) (4)



VI. List out types to traverse a Tree. Find out all type of traversal for following tree. (4)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